

Name: _____

Hammack

Score: _____

Directions: Do problems 1 and 3, and **one** of problems 2 or 4.

1. List the elements of the subgroup of $GL_2(\mathbb{R})$ generated by $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

2. Let p be a prime and take $r \in \mathbb{N}$. How many generators does $\mathbb{Z}_{p^r}$ have? Explain.

3. List all subgroups of $\mathbb{Z}_{18}$. Consider arranging them in a subgroup lattice.

4. Suppose $a, b \in G$, for some group G . Prove that ab and ba have the same order.